pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Changes in bp and Percent

WEEK 2 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

How far did it move?

Every morning note opens with the same line: where each series closed, and how far it moved on the day, the week, the month, and the year.

Unit

bp or percent?

Rows

A change from which earlier observation?

Holiday

A start date with no row at all

Make each decision once, in code, tested against Excel: change_bp, return_pct, change_since_bp.

Week 2: time series measures

Day 6	**Changes in bp and percent**
Day 7	Rolling windows, volatility
Day 8	Z-scores, percentile ranks
Day 9	Slopes and butterflies
Day 10	Case study: three years

Yields and spreads in bp; prices in percent

Series	Stored in	Its change	Formula
Treasury yield	percent	bp	(y_t - y_t-k) x 100
Credit spread	bp	bp	s_t - s_t-k
Bond price	per 100	percent	(P_t / P_t-k - 1) x 100

The last day in the file, by hand

10 Yr: 5.24 to 5.28 percent, a change of +4.0 bp

IG spread (synthetic): 98.0 to 97.0 bp, a change of -1.0 bp

The same 10 Yr change, unrounded

`4.0000000000000036`

Binary noise, the same in Excel. Round to display; compare with a tolerance.

k is the number of observations back: an earlier row of the market calendar, never a day with no row.

change_bp: rows back, whatever the gap

```
def change_bp(series, periods=1, percent_units=True):
    _check_series(series, "series")
    difference = series - series.shift(periods)
    return difference * 100 if percent_units else difference
```

percent_units

True: a yield in percent, times 100

False: a spread already in bp

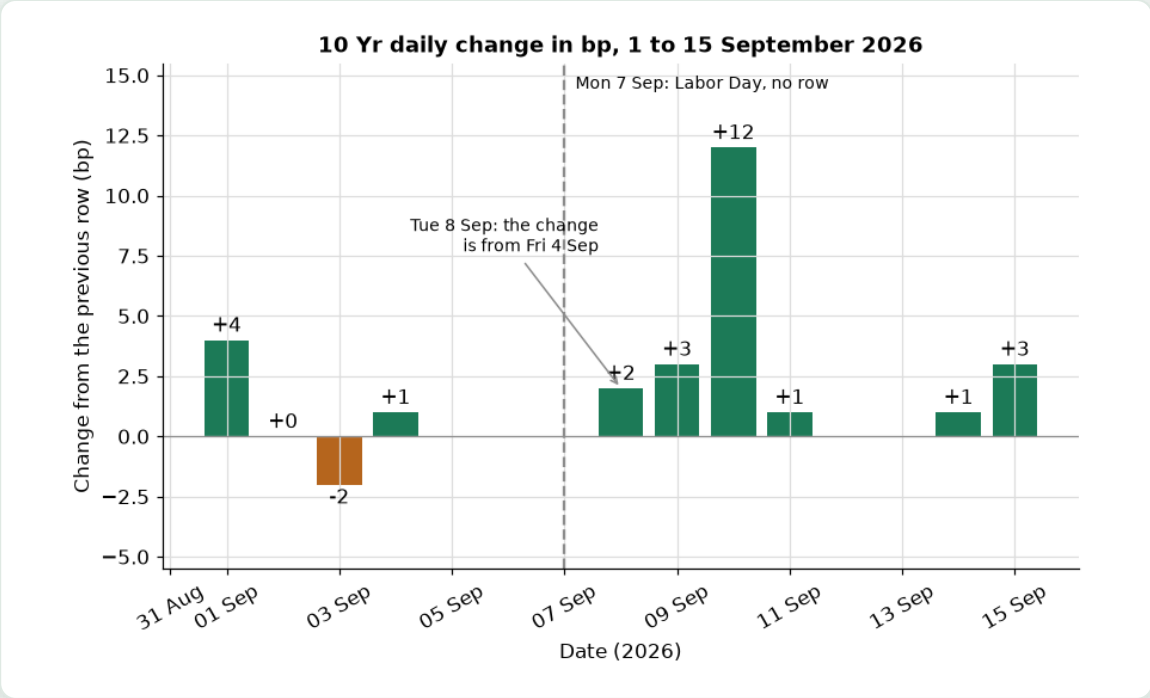
Left True on a spread: 100 times too big

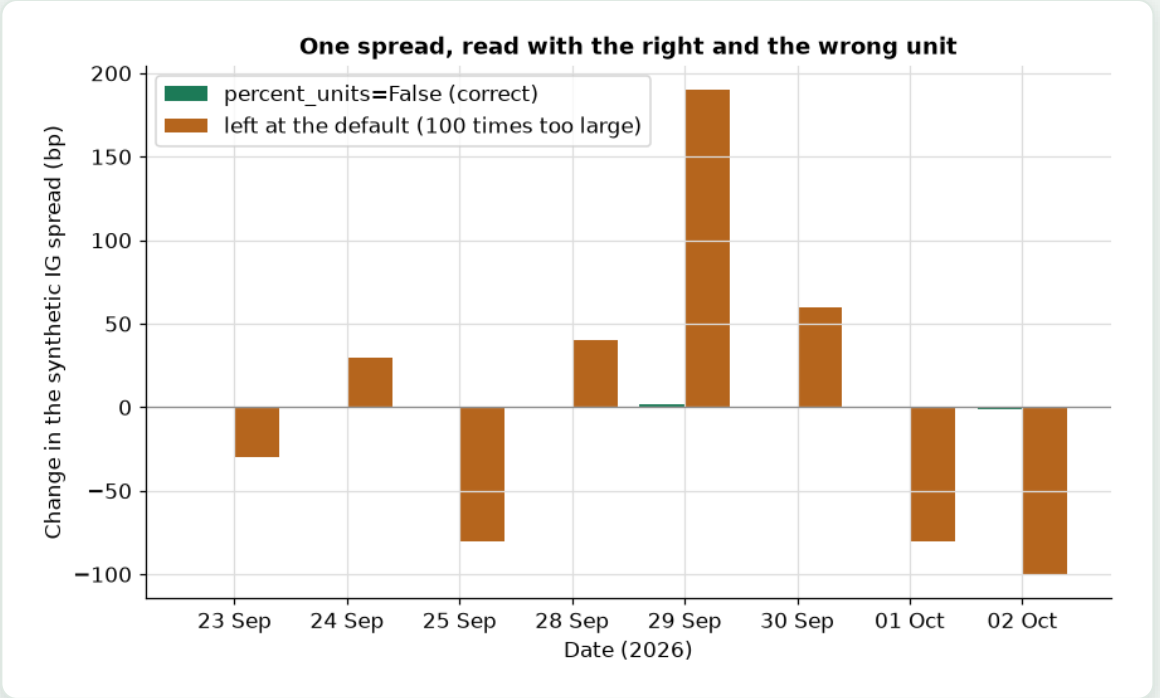
The week of Labor Day 2026: no row on Monday 7 September

2026-09-03 4.77	**2026-09-04 4.78**	2026-09-07 no row	**2026-09-08 4.80**	**2026-09-09 4.83**

Tuesday 2026-09-08 is a change from Friday: 4.78 to 4.80, +2 bp. A holiday is never a zero change.

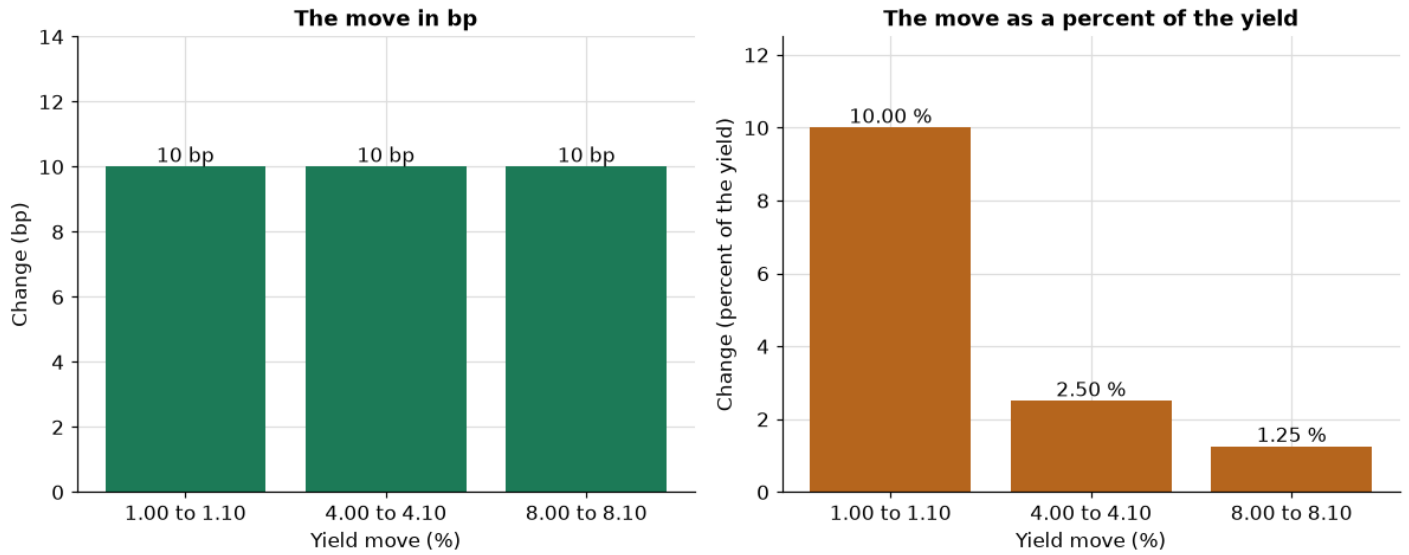
A holiday and a wrong unit, drawn





Left: Tuesday 8 September is a change from Friday. Right: a spread left at the yield default reads -100 for a -1.0 bp move.

Why a percent change of a yield misleads



Right for a price

pct_change() and =B3/B2-1 answer "what share of the price?" That is the question for a price.

Misleading for a yield

The same 10 bp move reads 10, 2.5, or 1.25 percent, depending only on the starting level.

Prices respond to the bp move (DV01 is price per 1 bp), so yields and spreads are compared in bp.

change_since_bp: a change since a date

Period	Since	Uses	10 Yr then (%)	Change (bp)
1 week	2026-09-25	2026-09-25	5.17	+11
1 month	2026-09-02	2026-09-02	4.79	+49
year to date	2025-12-31	2025-12-31	4.18	+110

As of 2026-10-02. Each end is an as-of value: the last observation on or before the date.

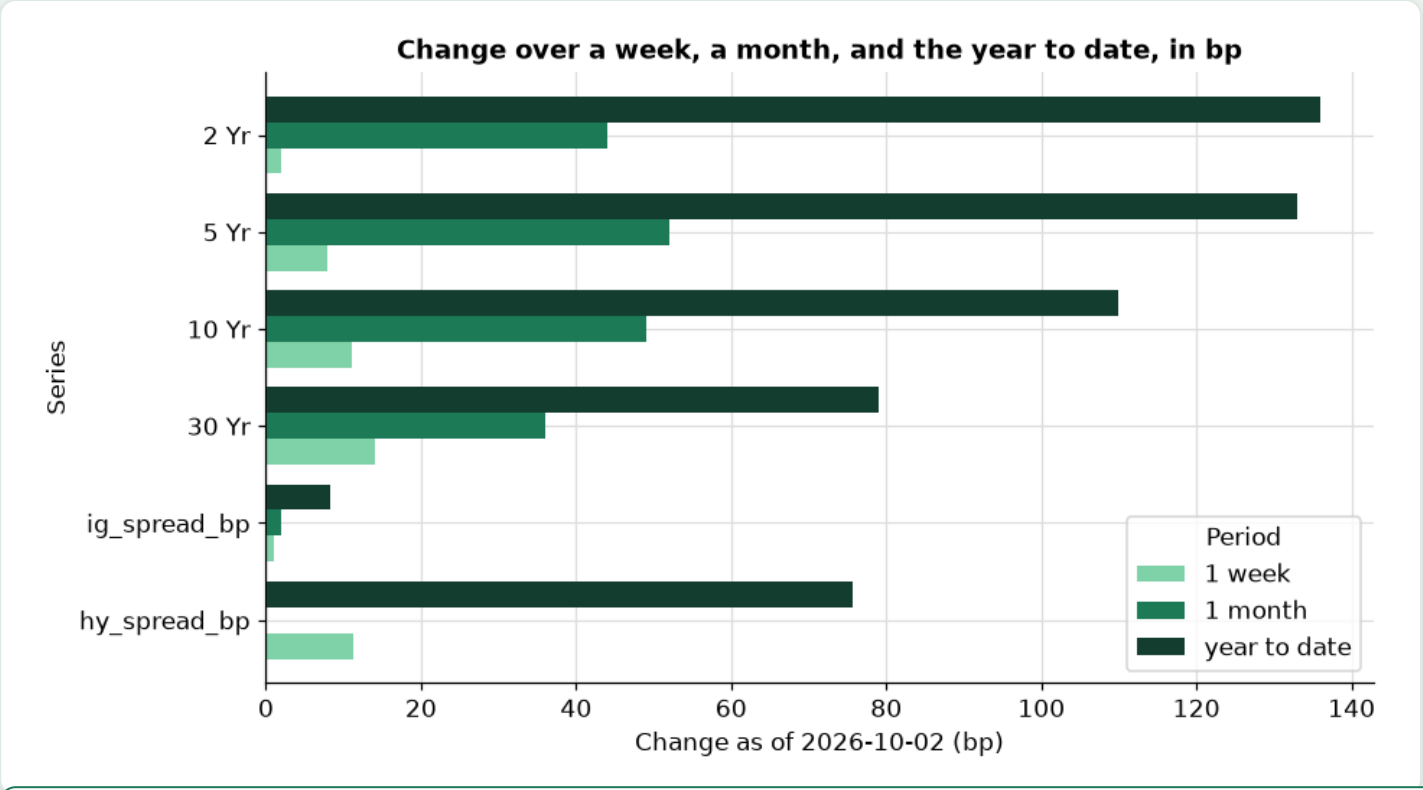
```
_, now = value_as_of(series, as_of)
_, then = value_as_of(series, since)
return (now - then) * 100
```

Since Labor Day, 2026-09-07

Uses Friday 2026-09-04: +51 bp to 30 September.

One week is 7 calendar days; one month is DateOffset(months=1), counted like EDATE.

The morning note, drawn



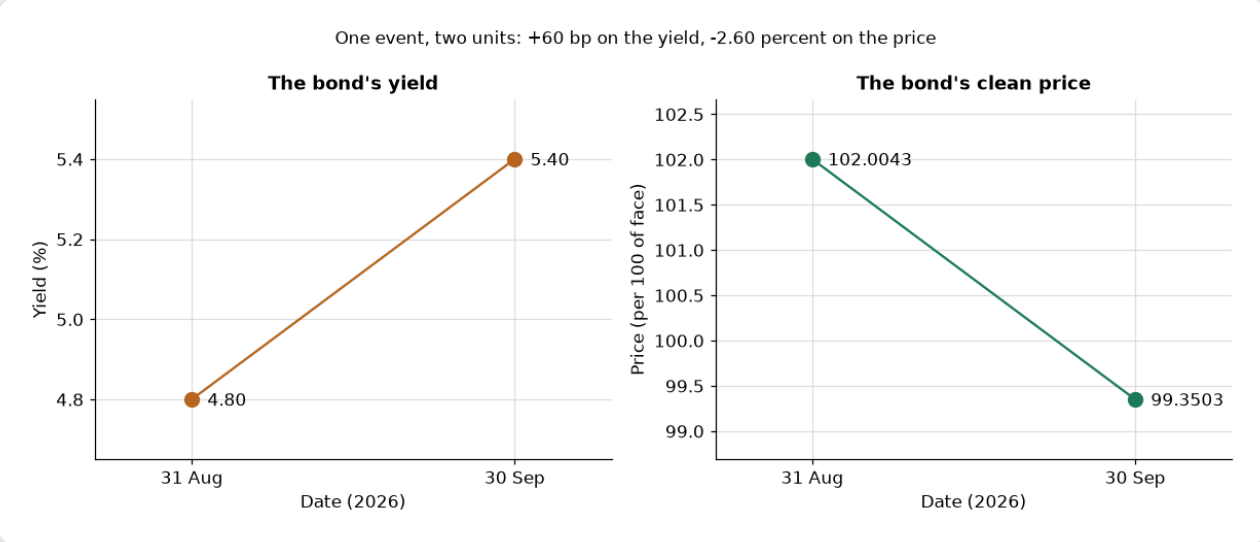
Series	1 week	1 month	YTD
2 Yr	+2.0	+44.0	+136.0
5 Yr	+8.0	+52.0	+133.0
10 Yr	+11.0	+49.0	+110.0
30 Yr	+14.0	+36.0	+79.0
ig_spread_bp	+1.1	+1.9	+8.2
hy_spread_bp	+11.2	+0.1	+75.7

Every number is in bp, as of 2 October 2026. The two spreads are synthetic.

Bp on one axis: a yield and a spread can be read across.

A price return from your bondmath

Quorvane Energy (synthetic, Course 2): 5.25 percent coupon, maturing 30 September 2031. Its yield moved by the 5 Yr's September change: 4.80 to 5.40 percent.



+60 bp

5 Yr over September: the yield change

-2.6018%

Clean price return: 102.004270 to 99.350327

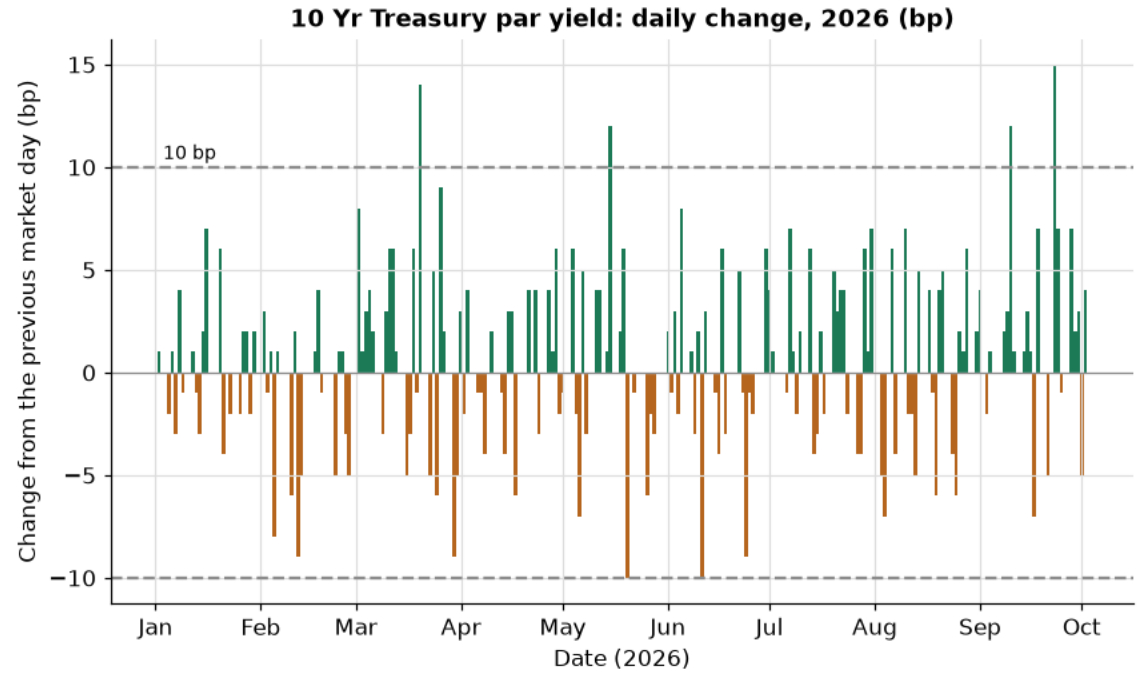
-2.6013%

Duration estimate: -4.3355 x 60 / 100

One event, two units. Duration is the bridge from a change in bp to a change in percent.

A clean price return: the coupon paid on 2026-09-30 and the month's accrued interest are not included.

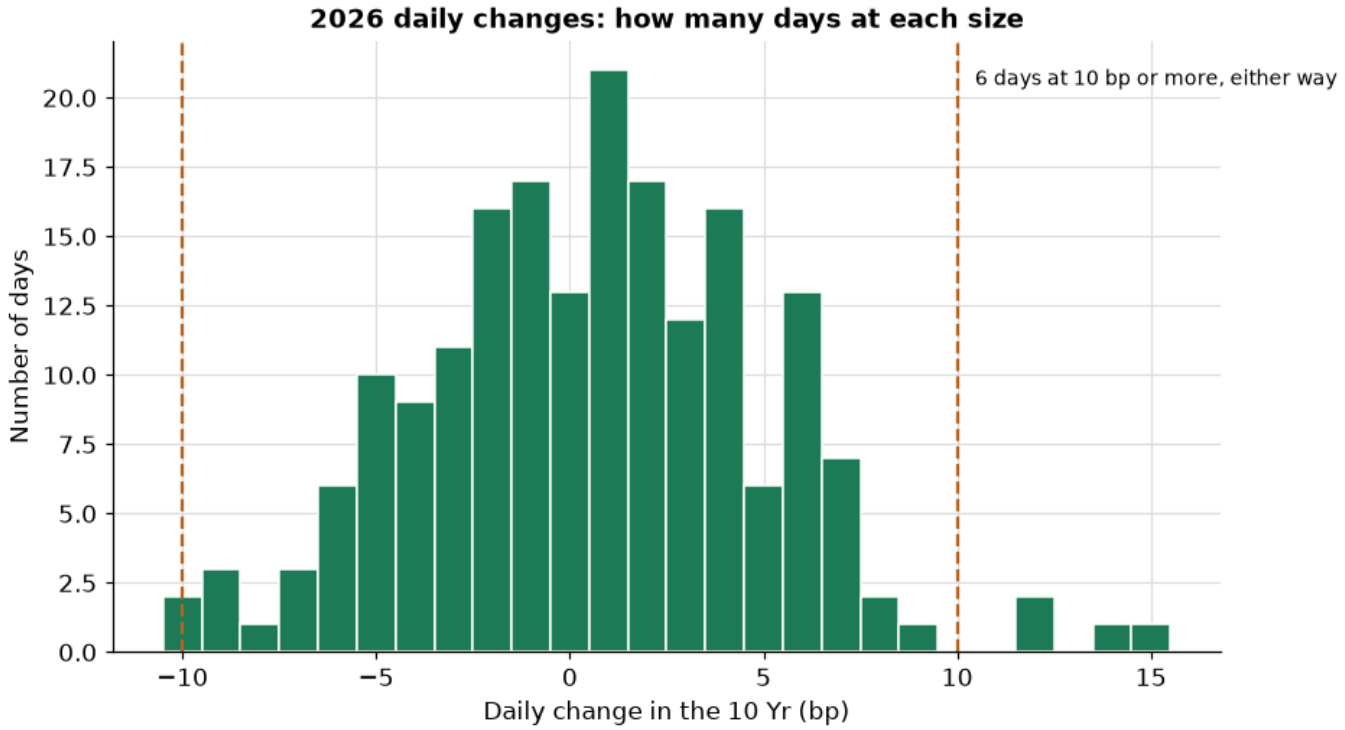
The largest daily moves of 2026



Date	Change (bp)
2026-09-23	15
2026-03-20	14
2026-05-15	12
2026-09-10	12
2026-05-20	-10
2026-06-11	-10

Days of 10 bp or more: 4 unrounded, 6 rounded. Two 10 bp falls are stored a hair short of 10.

How often: the 2026 changes by size



Bar chart or histogram

The bar chart answers when: each day in order. The histogram answers how often: each size counted.

6 days reached 10 bp either way, once the noise is rounded off.

Excel side by side

The daily change

```
=(B3-B2)*100
```

Filled down: equal to change_bp on all 190 rows of 2026.

=B2/B1-1 on 4.00 to 4.10 shows 2.5%: the percent trap.

A change since a date

```
=(B251-XLOOKUP(
  EDATE(A251,-1),
  A:A,B:B,,-1))*100
```

Match mode -1: the value on the date, or the last date before it. Equal on every row.

A price return

```
=PRICE(DATE(2026,8,31),
  DATE(2031,9,30),
  5.25/100,A1/100,100,2,0)
```

=(B2/B1-1)*100 returns -2.6018, as return_pct does.

Tested in Excel 16.0 build 20527, 2026-11-02.

Also tested: one week, year to date, a holiday start, and MAX with XLOOKUP for the date.

Save the day: git blame

Which day added each line? The history is rebuilt one commit a day, then today's commit goes on top.

```
git blame -s --root -L "/def get_secret/,+1" -L "/def value_as_of/,+1"
          -L "/def change_bp/,+1" marketdata.py
a45905d1  17) def get_secret(...
d3f9b277 251) def value_as_of(...
0ab3ed0c 339) def change_bp(...
```

git log --oneline: match the hashes

```
a45905d Day 1: APIs, Keys, and Secrets        d3f9b27 Day 4: Resampling and the As-Of...     0ab3ed0 Add change_bp, return_pct, and ...
```

get_secret came on day 1, value_as_of on day 4, change_bp today.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: daily_change_bp returns pct_change() * 100, labelled bp: 0.7634 for a 4 bp move.

Code that runs is not the same as code that is right.

What you can do now

- Yields and spreads change in bp; prices change in percent
- change_bp counts rows: a holiday is never a zero change
- change_since_bp: as-of values at both ends of a period
- Duration links a move in bp to a return in percent
- Round away binary noise before ranking; 25 tests pass

Week 2, Day 2 Rolling windows and volatility: daily changes in bp, windows in rows, and no lookahead